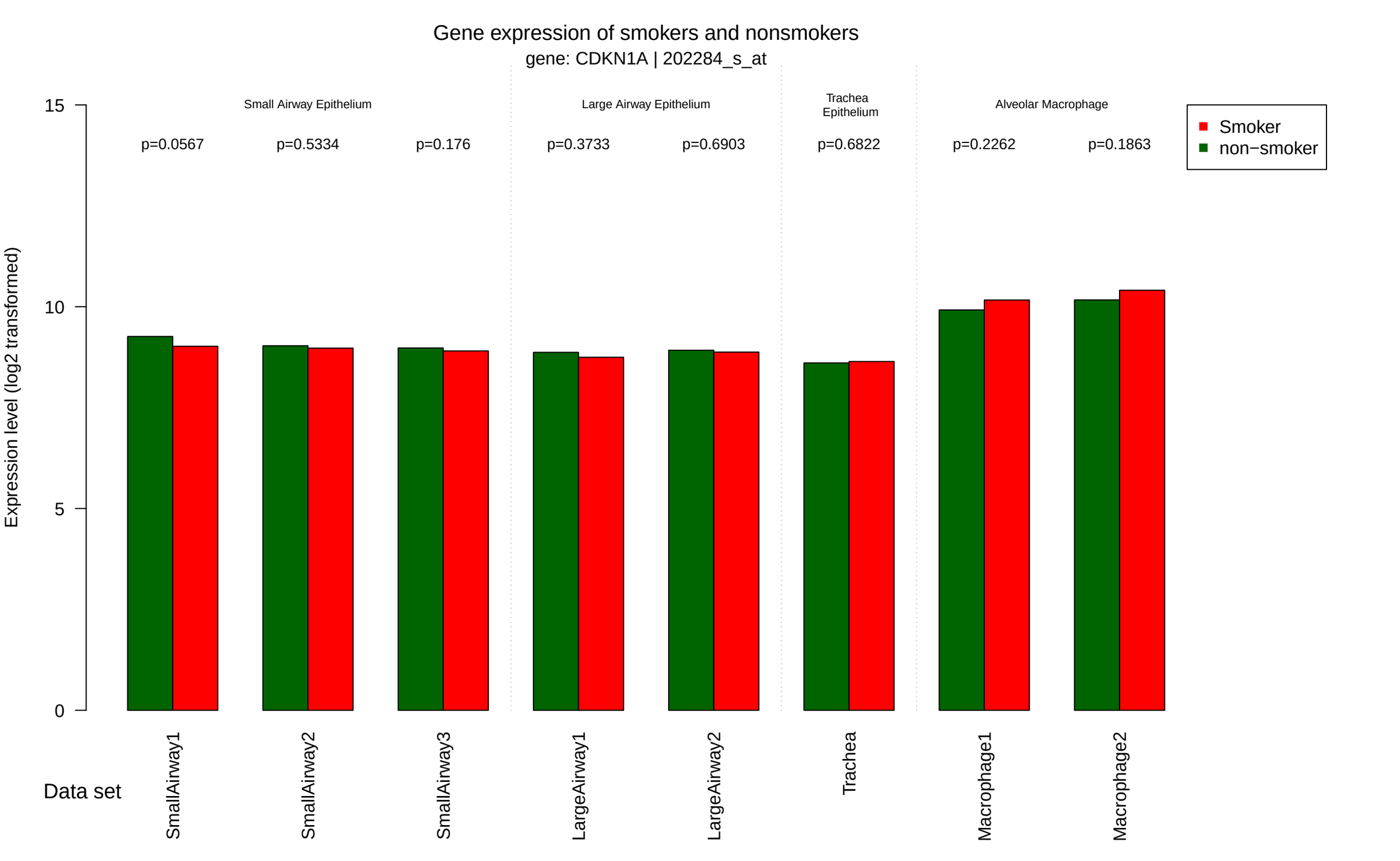
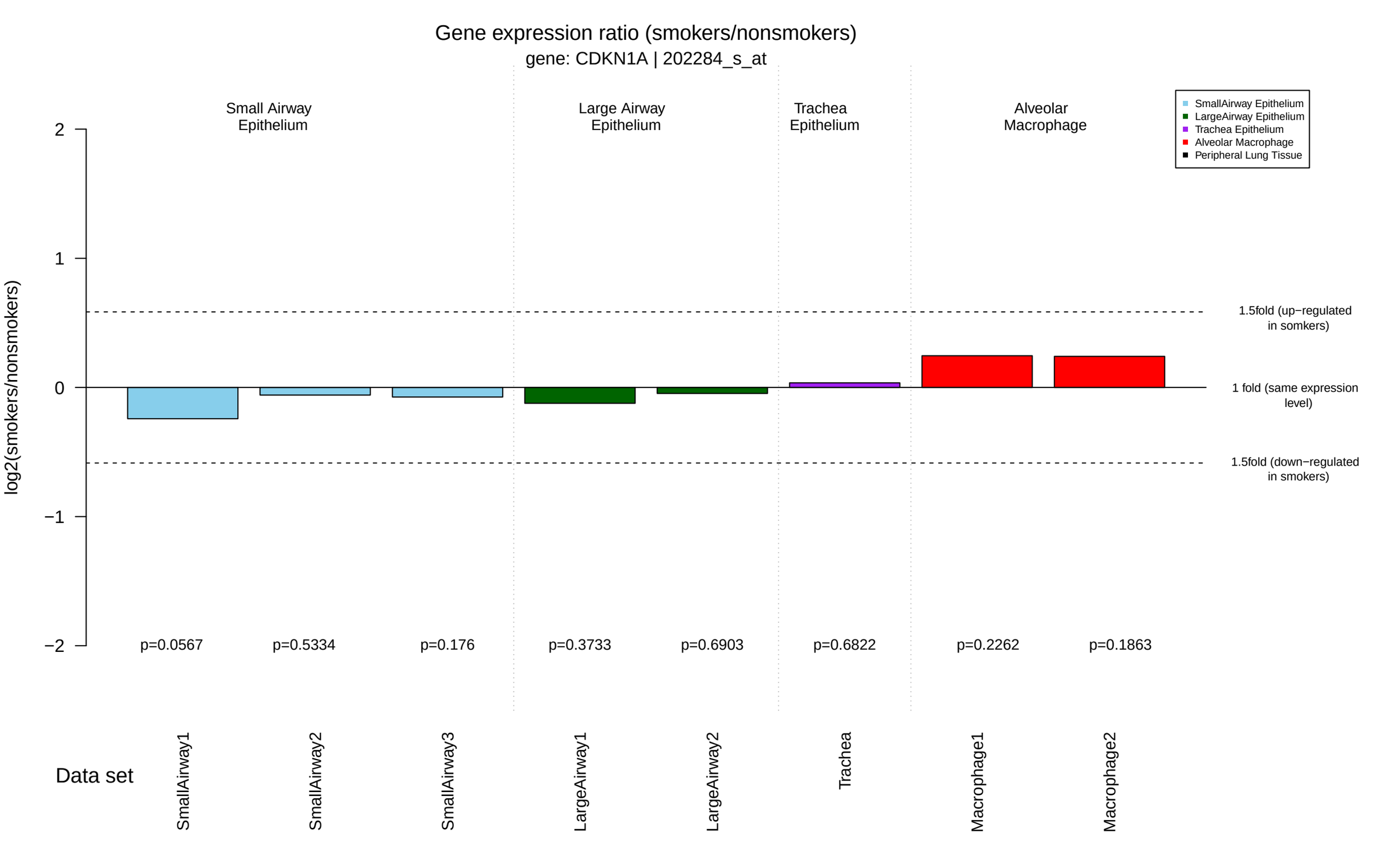
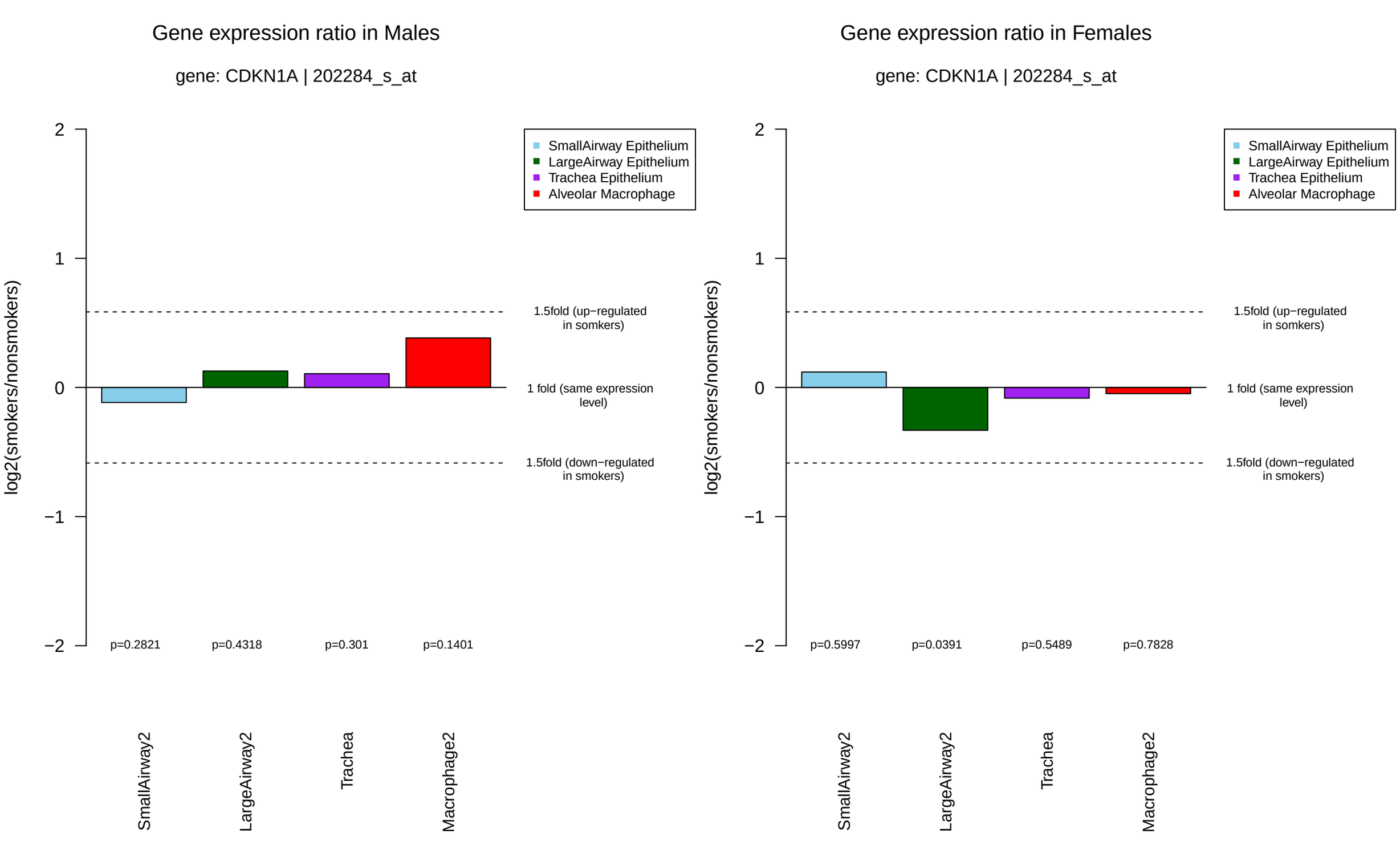
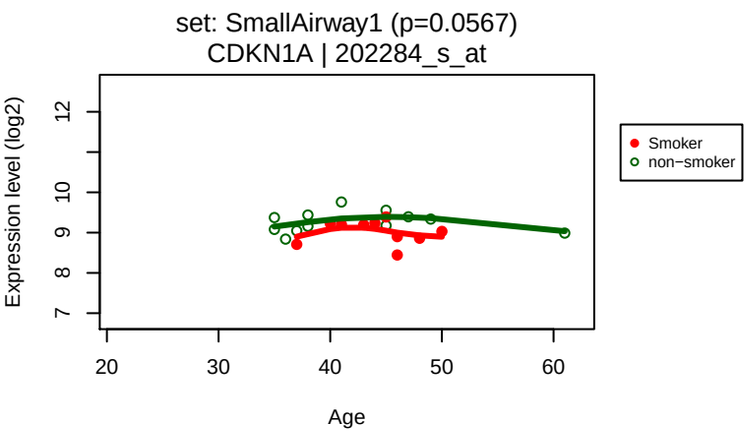
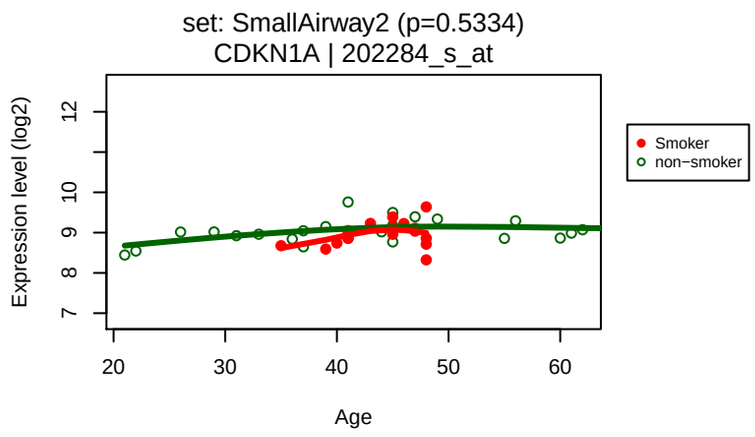
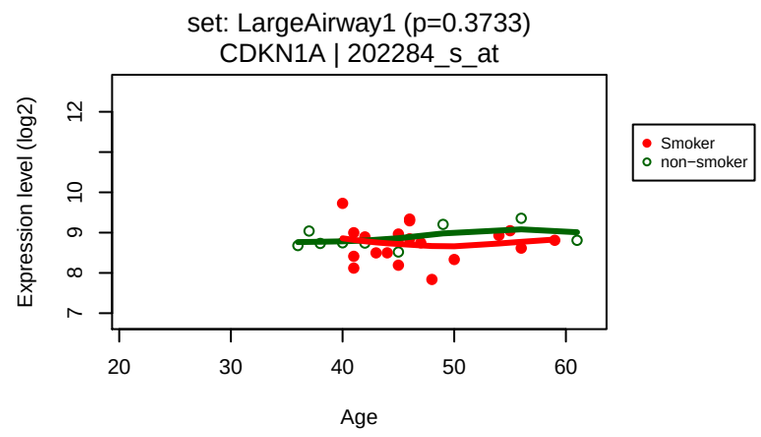

Gene expression of smokers and nonsmokers
gene: CDKN1A | 202284_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0567
p=0.5334
p=0.176
p=0.3733
p=0.6903
p=0.6822
p=0.2262
p=0.1863
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

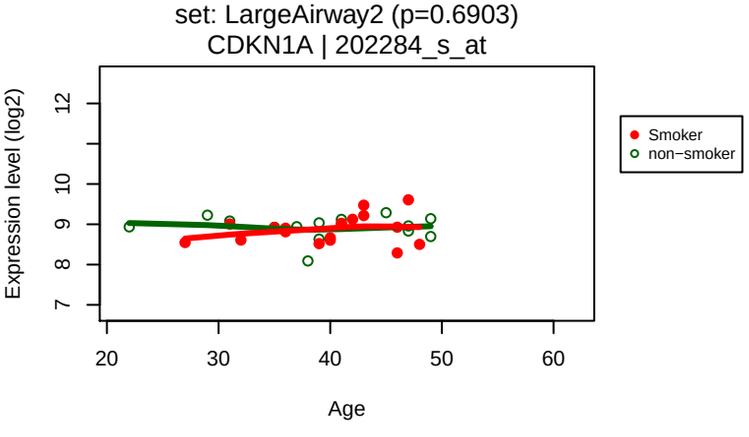

Gene expression ratio (smokers/nonsmokers)
gene: CDKN1A | 202284_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0567
p=0.5334
p=0.176
p=0.3733
p=0.6903
p=0.6822
p=0.2262
p=0.1863
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

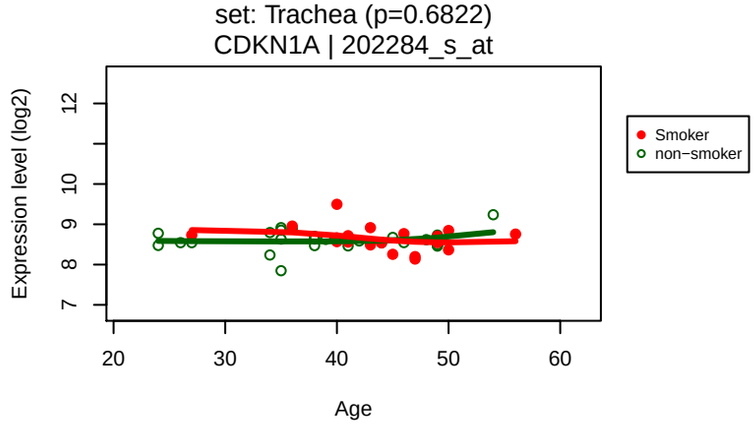

Gene expression ratio in Males
gene: CDKN1A | 202284_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2821
p=0.4318
p=0.301
p=0.1401
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CDKN1A | 202284_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5997
p=0.0391
p=0.5489
p=0.7828
SmallAirway2
LargeAirway2
Trachea
Macrophage2

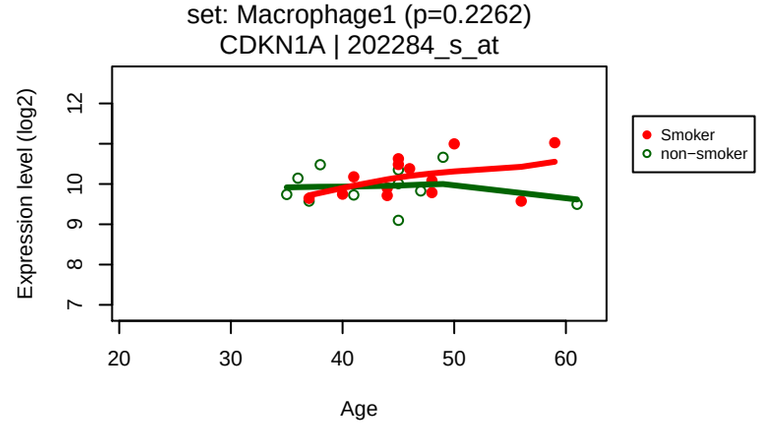

set: SmallAirway1 (p=0.0567)
CDKN1A | 202284_s_at
Expression level (log2)
Age
Smoker
non−smoker

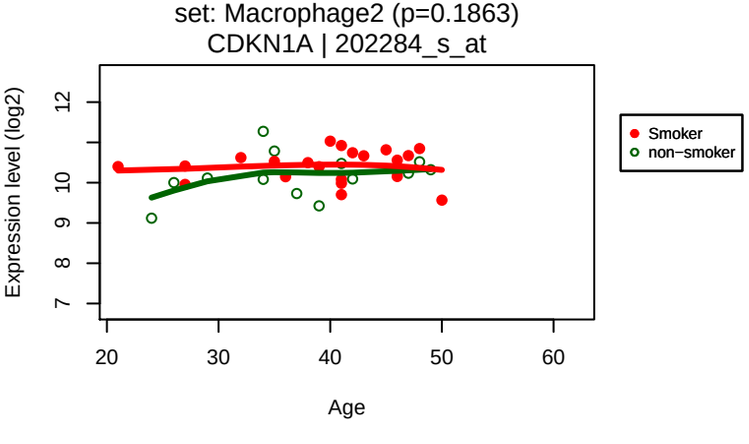

set: SmallAirway2 (p=0.5334)
CDKN1A | 202284_s_at
Expression level (log2)
Age
Smoker
non−smoker

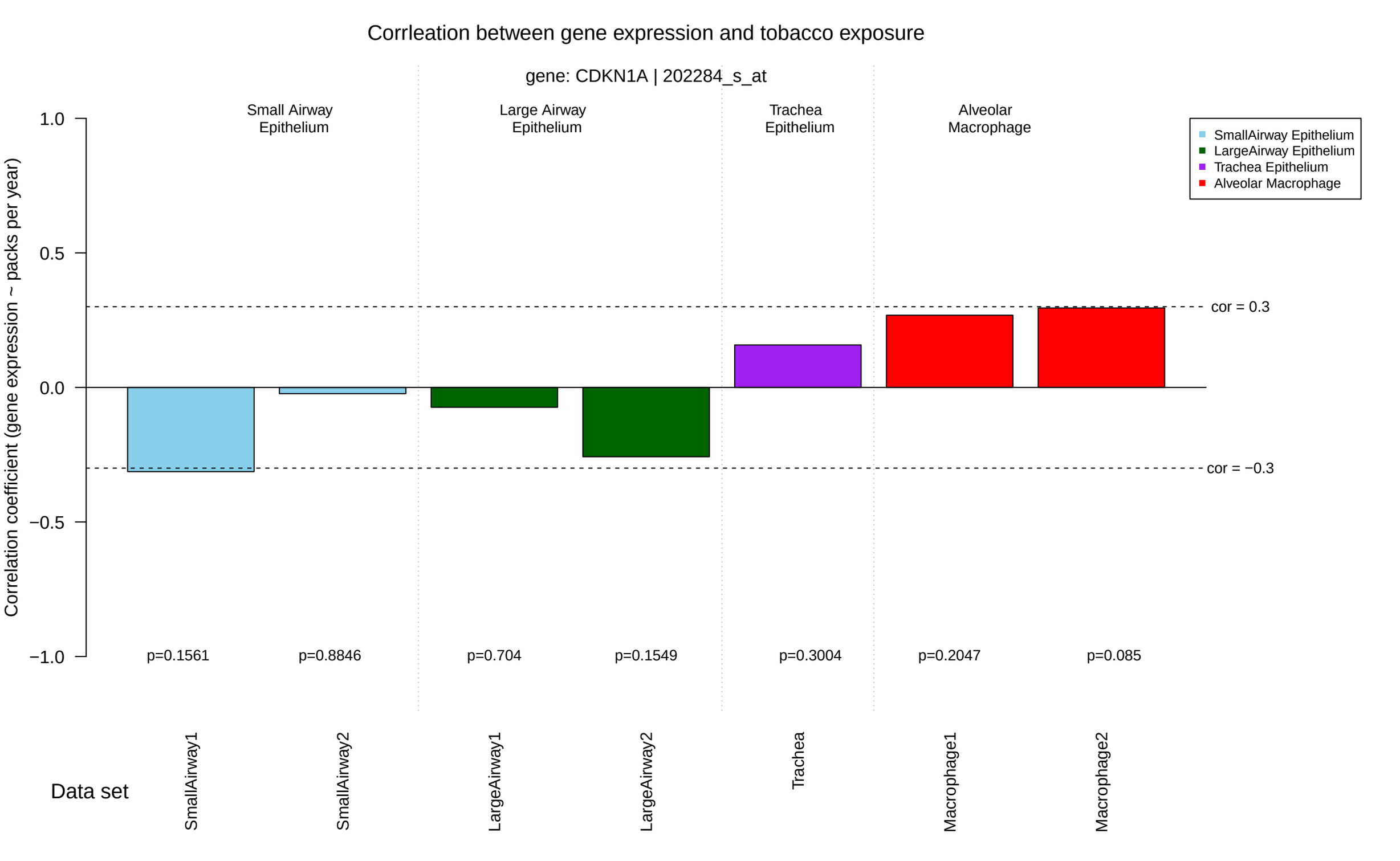

set: LargeAirway1 (p=0.3733)
CDKN1A | 202284_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6903)
CDKN1A | 202284_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6822)
CDKN1A | 202284_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2262)
CDKN1A | 202284_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1863)
CDKN1A | 202284_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CDKN1A | 202284_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1561
p=0.8846
p=0.704
p=0.1549
p=0.3004
p=0.2047
p=0.085
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2